Anuj Diwan

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Research Interests

My goal is to build speech and multimodal LLMs that understand and generate not just *what* is said, but *how*.
Major Interests: Speech & Omni-Modal LLMs, Post-Training, Expressive Speech, Data & Evaluation.
Other Interests: Interpretability, Multilingual Speech, Full-Duplex Dialogue, Joint Speech-Audio Generation.

Education

University of Texas at Austin — **Austin, TX**
PhD in Computer Science — *Aug 2021 - May 2027 (expected)*
Concentration: Speech and Natural Language Processing
GPA 4.00/4.00

Indian Institute of Technology Bombay — **Mumbai, India**
B.Tech in Computer Science and Engineering with Honors — *Jul 2017 - Aug 2021*
Minor in Applied Statistics and Informatics
B.Tech CPI: 9.75/10, Minor CPI: 10/10

Research Experience

UT Austin Computer Science — **Austin, TX**
PhD Student, Advisors: Prof. David Harwath, Prof. Eunsol Choi — *Aug 2021 - May 2027 (expected)*

- **ParaSpeechCaps (EMNLP'25):** 2,400+ h of crowdsourced and LLM-annotated speech, 59 style tags; +7.9% consistency and +15.5% naturalness MOS in style-prompted TTS; 16K+ Hugging Face downloads, 166 GitHub stars.
- **ParaSpeechCLAP (Interspeech'26, Best Student Paper shortlist):** Dual-encoder speech-text style embeddings; 88.8 vs 72.6 R@10 over ParaCLAP fine-tuned on the same data.
- **Interpretability:** Interpretable acoustic feature discovery in autoregressive voice-cloning TTS models (under review).
- **Earlier work:** Textless speech-to-speech translation from 20–60 h of parallel data (EMNLP'24 Findings); efficient-attention profiling (ACL'23); visuolinguistic compositionality (EMNLP'22 Oral); zero-shot video moment retrieval.
- **Collaborations:** Task editor for the 180-task Dynamic-SUPERB, reviewing 23 task proposals (ICLR'25); human evaluations for VoiceCraft-X, 11-language TTS (EMNLP'25); data creation for code-switched speech datasets covering 113 language pairs and 313 h of in-the-wild speech (Interspeech'25, LREC'26).

Meta Superintelligence Labs — **New York, NY**
Research Scientist Intern, Mentors: Kartik Audhkhasi, Bhuvana Ramabhadran — *May - Oct 2026*

- Cross-team work with FAIR: post-training conversational speech LLMs to ground responses in user paralinguistics via on-policy self-distillation; instruct model matches or beats the off-the-shelf thinking model (paper under review).

Google DeepMind — **Mountain View, CA**
Student Researcher, Mentors: Yu Zhang, Ankur Bapna — *May - Dec 2023*

- Improving out-of-domain generalization for TTS and speech translation via speech LLM chain-of-thought prompting.

Meta FAIR — **Seattle, WA**
Research Intern (AI), Mentors: Abdelrahman Mohamed, Wei-Ning Hsu — *May - Dec 2022*

- Benchmarks and models for parameter-efficient continual learning for speaker adaptation; 20.65% relative WER reduction over parameter-matched baselines while tuning <13% of parameters (ICASSP'23).

Undergraduate Research
IIT Bombay, Adobe Research, UCLouvain — *2019 - 2021*

- **IIT Bombay** (Preethi Jyothi, Sunita Sarawagi): Low-resource and code-switched ASR (3 Interspeech'21 papers); co-organized the MUCS 2021 challenge with ∼600 h of speech in 7 Indian languages and Kaldi/ESPnet baselines.
- **Adobe Research** (Summer 2020): RL-based scheduling of ML jobs on interruptible cloud VMs (US patent).
- **UCLouvain** (Summer 2019, Pierre-Antoine Absil): Riemannian optimization for matrix completion.

Publications/Patents (* equal contribution)

2026 **Anuj Diwan**, Puyuan Peng, Haohe Liu, Abdelrahman Mohamed, Bhuvana Ramabhadran, Kartik Audhkhasi. *Teaching Speech Assistants to Act on Vocal Cues*. **Under review**.

2026 **Anuj Diwan**, David Harwath, Eunsol Choi. *Discovering Interpretable Acoustic Features in Autoregressive Voice-Cloning TTS Models*. **Under review**.

2026 **Anuj Diwan**, Eunsol Choi, David Harwath. *ParaSpeechCLAP: A Dual-Encoder Speech-Text Model for Rich Stylistic Language-Audio Pretraining*. **Interspeech'26**. 🏆 **Best Student Paper Shortlist**. **[code]**

2026 Brian Yan, Qingzheng Wang, Matthew Wiesner, **Anuj Diwan**, et al. *CS-YODAS: A Mined Dataset of In-the-Wild Code-Switched Speech*. **LREC'26**. **[dataset]**

2025 **Anuj Diwan**, Zhisheng Zheng, David Harwath, Eunsol Choi. *Scaling Rich Style-Prompted Text-to-Speech Datasets*. **EMNLP'25**. **[code][demo][dataset]**

2025 Zhisheng Zheng, Puyuan Peng, **Anuj Diwan**, et al. *VoiceCraft-X: Unifying Multilingual, Voice-Cloning Speech Synthesis and Speech Editing*. **EMNLP'25**. **[code][demo]**

2025 Younghan Park, **Anuj Diwan**, David Harwath, Eunsol Choi. *Rhapsody: A Dataset for Highlight Detection in Podcasts*. **COLM'25**.

2025 Brian Yan, Injy Hamed, Shuichiro Shimizu, . . . , **Anuj Diwan**, et al. *CS-FLEURS: A Massively Multilingual and Code-Switched Speech Dataset*. **Interspeech'25**.

2025 Chien-yu Huang, . . . , **Anuj Diwan**, et al. *Dynamic-SUPERB Phase-2: A Collaboratively Expanding Benchmark for Measuring the Capabilities of Spoken Language Models with 180 Tasks*. **ICLR'25**.

2024 **Anuj Diwan**, Anirudh Srinivasan, David Harwath, Eunsol Choi. *Textless Speech-to-Speech Translation With Limited Parallel Data*. **EMNLP'24 Findings**.

2023 **Anuj Diwan**, Eunsol Choi, David Harwath. *When to Use Efficient Self Attention? Profiling Text, Speech and Image Transformer Variants*. **ACL'23**.

2023 **Anuj Diwan**, Ching-Feng Yeh, Wei-Ning Hsu, Paden Tomasello, Eunsol Choi, David Harwath, Abdelrahman Mohamed. *Continual Learning for On-Device Speech Recognition using Disentangled Conformers*. **ICASSP'23**.

2022 **Anuj Diwan***, Layne Berry*, Eunsol Choi, David Harwath, Kyle Mahowald. *Why is Winoground Hard? Investigating Failures in Visuolinguistic Compositionality*. **EMNLP'22 (Oral)**. **[code]**

2022 **Anuj Diwan***, Puyuan Peng*, Raymond J. Mooney. *Zero-shot Video Moment Retrieval With Off-the-Shelf Models*. **TL4NLP Workshop @ NeurIPS'22**.

2021 **Anuj Diwan**, Preethi Jyothi. *Reduce and Reconstruct: ASR for Low-Resource Phonetic Languages*. **Interspeech'21**. 🏆 **Best Student Paper Shortlist**.

2021 **Anuj Diwan**, Rakesh Vaideeswaran, Sanket Shah, Ankita Singh, et al. *MUCS 2021: Multilingual and Code-Switching ASR Challenges for Low Resource Indian Languages*. **Interspeech'21**. **[code]**

2021 Shreya Khare*, Ashish Mittal*, **Anuj Diwan***, Sunita Sarawagi, Preethi Jyothi, Samarth Bharadwaj. *Low Resource ASR: The Surprising Effectiveness of High Resource Transliteration*. **Interspeech'21**.

2024 Subrata Mitra, Sunav Choudhary, Shaddy Garg, **Anuj Diwan**, et al. *Scheduling and Control of Executable Jobs over Compute Instances*. **US Patent 12,014,217 B2** (Adobe).

Honors & Awards

- ISCA Best Student Paper shortlist at Interspeech 2026 (ParaSpeechCLAP) and 2021 (Reduce & Reconstruct).
- Awarded the Excellence in Research Award 2021 by the IIT Bombay CS Department.
- IIT Bombay: Institute Academic Prize (2017-18), Excellence in TA Award, top-1% grades in 7 courses.
- Among top 35 in India selected for the International Mathematical Olympiad Training Camp in 2016.
- Achieved All India Rank 118 among ∼160k candidates in JEE Advanced 2017.

Invited Talks

Toyota Technological Institute at Chicago
"Scaling Rich Style-Prompted Text-to-Speech Datasets" *Sep 2025*

AI4Bharat, IIT Madras
"Textless Speech-to-Speech Translation With Limited Parallel Data" *Jul 2024*

Service

Reviewing: ICLR, ACL, EMNLP, ARR, Interspeech, ICASSP, IEEE TASLP, IEEE JSTSP, IEEE OJSP.

Teaching: TA at UT Austin for Intro to Speech and Audio Processing (Prof. David Harwath), Spring 2024. TA at IIT Bombay for Automatic Speech Recognition (Prof. Preethi Jyothi), Spring 2021.

Mentoring: Younghan Park (project led to Rhapsody, published at COLM'25); Rishi Salem (completed an undergraduate research project on speech negotiation models).